

Exact One-Electron Wilson Qualification in Aion

Physics, implementation, numerical evidence, and artifact index

Aion development record

19 September 2026

Qualification status. Gates G0–G8 were explicitly reviewed and accepted by the user. The accepted scope is fixed-centre, linear, one-electron straight-Wilson dynamics and the declared P0, E1, gB1, B1, and C1 truncations on the tabulated fixtures. The selected post-G8 branch is the Aion adiabatic pure-LDA/GGA TDDFT closure. That nonlinear closure is a future branch and is not part of the evidence summarized here.

Repository state represented by this report:

feature/exact-one-electron-
wp7-action-observables

The record begins from the accepted WP8 synthesis at commit 2fa0214 and the subsequent review record. Raw executions are immutable and retain their execution-time status; acceptance is carried only by the separate files in `docs/reviews`.

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1 Purpose and how to use this report

This document is the principal entry point to the exact one-electron Wilson qualification campaign. It has four jobs:

1. state the physical and mathematical problem that was actually tested;
2. describe the implemented evaluator, time connection, propagator, and action-derived observables;
3. summarize the numerical results and the failed as well as accepted algorithmic explorations; and
4. index the source code, plans, review decisions, raw executions, derived tables, figures, and verification tools needed to audit or reproduce every conclusion.

The report deliberately distinguishes four kinds of statement:

Mathematical identity

A consequence of the declared continuum action, Wilson frame, and index conventions. An identity may be true even before code exists.

Implemented feature

A reusable Aion code path. Its existence alone does not show that it was executed in the accepted environment.

Executed numerical evidence

A hash-authenticated run or derived analysis tied to a repository commit, fixture, backend, precision, grid, and timestep.

Reviewed evidence

An explicit user decision in `docs/reviews`. Only this level changes a gate to accepted. Raw manifests remain `executed_unreviewed`.

This distinction is especially important for WP6: the first time integrator was implemented and executed but failed the declared model-difference stability criterion. The later mixed-index Gauss–Magnus experiment was then implemented, executed to 128 intervals in the hard case, and accepted.

2 Executive scientific conclusions

1. The direct dressed-orbital and endpoint-factorized routes reproduce the same exact straight-Wilson one-electron matrices at the measured quadrature floor. Gauge-representative changes, gauge-origin changes, field reversal, pair reversal, and rigid rotation behave as required.
2. P0 is a controlled small-field approximation only inside explicit system, basis, geometry, and operator brackets. No molecule-independent P0 cutoff is supported by the data.
3. The triangle-flux correction gB1 improves the overlap metric but does not supply the anchored-vector response required by the kinetic, nuclear-attraction, and total mechanical channels. Full B1 resolves both onsite and offsite mechanical blocks. Across the classified blocks, the median B1 error-reduction factor is 183.10.
4. Multicentre loop flux makes the missing channels spectrally visible. In the cc-pVDZ above-threshold fixtures, P0 and gB1 can shift generalized eigenvalues by order-one to tens of hartree, whereas B1/C1 reduce the shift to roughly 3×10^{-2} – 4×10^{-2} Ha.
5. The natural mixed-index density $D = PS$ obeys an orthogonal-looking commutator equation even though the AO frame is nonorthogonal and moving. A fourth-order two-node Gauss–Magnus link with a [2/2] Padé exponential propagates it by similarity, without Löwdin orthogonalization, endpoint Cholesky correction, or metric projection.
6. The accepted 32/64/128-interval hard-case continuation measures global orders from 3.707 to 3.990. The largest 64-to-128 change in a model-minus-exact density difference is 0.004454 of the final difference, below the declared 0.25 limit.
7. In the distorted one-electron H_3^{2+} above-threshold pulse, only the complete first-order model C1 closely tracks the exact endpoint observables. Its absorbed-energy difference is 2.2166×10^{-7}

Ha, compared with a timestep difference uncertainty of 7.95×10^{-14} Ha; the residual is therefore a resolved model truncation difference, not a propagation artifact.

8. Exact action derivatives give site charge, oriented-link current, weak continuum-current pairings, Ward identities, continuity, and uniform electric power. On the finest accepted electric trajectory the integrated work defect is -1.236×10^{-9} Ha and the instantaneous action identity closes at 4.88×10^{-16} .
9. CPU and physical-CuPy trajectories agree at approximately 1.5×10^{-15} – 4.1×10^{-15} in the WP7 parity comparison.

3 Physical scope and terminology

3.1 Included

The qualified problem uses finite molecules with fixed nuclei and fixed AO anchors; real spherical Gaussian AOs from PySCF; straight paths from each anchor to the electronic coordinate; local all-electron nuclear attraction; prescribed electromagnetic sources; and no Maxwell backreaction. Static tests use uniform magnetic fields. Dynamic tests use uniform electric fields and a Maxwell-consistent time-dependent uniform magnetic field with its affine induction electric field.

The pair campaign covers H–H, O–H, N₂, and CO; STO-3G, cc-pVDZ, and aug-cc-pVDZ; bond-length scales 0.8, 1.0, and 1.25; and parallel, perpendicular- x , perpendicular- y , and oblique field directions. The multicentre and dynamic fixtures use equilateral and distorted three-proton frameworks. Their electronic problem has one electron, so it is H_3^{2+} and is a geometric qualification fixture, not a claim about physical neutral H_3 or two-electron H_3^+ .

3.2 Excluded

The accepted evidence does not cover self-consistent Hartree or XC feedback, nonlinear SCEM propagation, moving nuclei or anchors, Pulay forces, pseudopotentials, periodic systems, general nonuniform-field power balance, Maxwell backreaction, or pointwise reconstruction of the continuum current density. Weak current pairings are source-covector tests, not a pointwise current field.

3.3 Meaning of “exact”

“EX” or “exact Wilson” means the analytic pullback of the declared one-electron action to the finite subspace spanned by the chosen straight-Wilson-dressed Gaussian orbitals. It is exact with respect to that restriction, while retaining finite-basis and measured real-space quadrature errors. It is not an exact continuum Schrödinger solution.

4 Wilson frame, restricted action, and approximation hierarchy

4.1 Electromagnetic and AO conventions

In atomic units the electron has $q = -1$, $m = 1$, and $\hbar = 1$ in the executed campaigns. The covariant derivatives and physical fields are

$$D_t^\Phi = \partial_t + \frac{iq}{\hbar}\Phi, \quad \pi_{\mathbf{A}} = -i\hbar\nabla - q\mathbf{A}, \quad (1)$$

$$\mathbf{E} = -\partial_t\mathbf{A} - \nabla\Phi, \quad \mathbf{B} = \nabla \times \mathbf{A}. \quad (2)$$

The scalar-potential energy is carried by D_t^Φ and is not added again to the mechanical energy.

For a bare real AO e_μ anchored at $\mathbf{R}_\mu$, define the straight-path phase, Wilson factor, and dressed AO by

$$a_\mu[\mathbf{A}](\mathbf{r}, t) = \int_0^1 (\mathbf{r} - \mathbf{R}_\mu) \cdot \mathbf{A}(\mathbf{R}_\mu + s(\mathbf{r} - \mathbf{R}_\mu), t) ds, \quad (3)$$

$$W_\mu = \exp\left(\frac{iq}{\hbar} a_\mu\right), \quad \chi_\mu = W_\mu e_\mu. \quad (4)$$

The coefficient density is $P = CfC^\dagger$ and lower matrices contract as $\langle O \rangle = \text{Re Tr}(PO)$. No extra closed-shell factor is inserted.

4.2 Metric, connection, mechanical matrix, and equation of motion

The exact projected metric, temporal connection, and one-electron mechanical matrix are

$$S_{\mu\nu} = \langle \chi_\mu | \chi_\nu \rangle, \quad (5)$$

$$(\omega_t)_{\mu\nu} = \langle \chi_\mu | D_t^\Phi \chi_\nu \rangle, \quad (6)$$

$$K_{\mu\nu} = \frac{1}{2m} \langle \pi_{\mathbf{A}} \chi_\mu | \pi_{\mathbf{A}} \chi_\nu \rangle + \langle \chi_\mu | v_{\text{nuc}} | \chi_\nu \rangle. \quad (7)$$

They satisfy metric compatibility,

$$\dot{S} = \omega_t + \omega_t^\dagger. \quad (8)$$

The restricted real action yields

$$\boxed{i\hbar (S\dot{C} + \omega_t C) = KC}. \quad (9)$$

The ordinary-derivative matrix $K - i\hbar\omega_t$ is not the mechanical energy. This distinction prevents connection energy from being mislabelled as an absorbed mechanical energy.

4.3 Endpoint link and internal finite-spread geometry

For the centre path from ket anchor $\mathbf{R}_\nu$ to bra anchor $\mathbf{R}_\mu$,

$$\Theta_{\mu\nu} = \exp\left[\frac{iq}{\hbar} \int_{\mathbf{R}_\nu}^{\mathbf{R}_\mu} \mathbf{A} \cdot d\ell\right], \quad (10)$$

$$F_{\mu\nu}^B(\mathbf{r}) = \Theta_{\mu\nu}^{-1} W_\mu(\mathbf{r})^* W_\nu(\mathbf{r}). \quad (11)$$

Θ is an exact gauge-covariant endpoint link. F^B is the gauge-invariant holonomy through the triangle formed by the two anchors and $\mathbf{r}$. With the anchored magnetic residual vector

$$\mathbf{C}_\mu^B(\mathbf{r}) = \int_0^1 s(\mathbf{r} - \mathbf{R}_\mu) \times \mathbf{B}(\mathbf{R}_\mu + s(\mathbf{r} - \mathbf{R}_\mu)) ds, \quad (12)$$

one has $\pi_{\mathbf{A}} \chi_\mu = W_\mu(-i\hbar\nabla + q\mathbf{C}_\mu^B)e_\mu$. For a uniform $\mathbf{B}$, both F^B and $\mathbf{C}_\mu^B$ are analytic affine geometry factors. Full matrices factor as the endpoint link times an endpoint-removed internal amplitude, for example $S = \Theta \odot \bar{S}$ and $K = \Theta \odot \bar{K}$.

The exact temporal connection follows from the electric line integral

$$\mathcal{V}_\nu^E(\mathbf{r}, t) = \int_0^1 (\mathbf{r} - \mathbf{R}_\nu) \cdot \mathbf{E}(\mathbf{R}_\nu + s(\mathbf{r} - \mathbf{R}_\nu), t) ds, \quad (13)$$

as

$$(\omega_t)_{\mu\nu} = \frac{iq}{\hbar} \Theta_{\mu\nu} \int \bar{N}_{\mu\nu}(\mathbf{r}, t) [\Phi(\mathbf{R}_\nu, t) - \mathcal{V}_\nu^E(\mathbf{r}, t)] d^3r. \quad (14)$$

Thus the connection contains the anchor-scalar sector, endpoint-link motion, and the internal electric finite-spread response.

4.4 Comparison models

Table 1 gives the implemented action-consistent hierarchy. “gB1” means the geometric first magnetic correction only; “B1” includes the full first magnetic mechanical response; and “C1” is the complete simultaneous first internal-amplitude order.

Table 1: Action-level matrix content of each comparison model.

Model	Metric	Mechanical matrix	Temporal connection
EX	exact	exact	exact
P0	endpoint link $\times$ bare metric	endpoint link $\times$ bare mechanical matrix	P0-compatible connection
E1	P0	P0	P0 plus internal electric increment
gB1	full first magnetic metric	P0	metric-compatible magnetic connection
B1	full first magnetic metric	full first magnetic mechanical matrix	metric-compatible magnetic connection
C1	full first magnetic metric	full first magnetic mechanical matrix	B1 connection plus internal electric increment

In the static electric-free sector C1 equals B1. At zero magnetic curvature and for a spatially uniform electric field, the exact connection reduces to

$$\omega_t = \omega_{P0} + \frac{i}{\hbar} V_{E1}, \quad V_{E1} = - \sum_{\alpha} E_{\alpha} D_{\alpha}, \quad (15)$$

so the existing P0+E1 formulation is recovered in its declared domain. E1 belongs in the connection; it is not an additional mechanical-energy term.

5 Implementation architecture

5.1 How PySCF is used

PySCF is the analytic and real-space electronic-structure backend, but it does not determine the Wilson physics. Aion uses it in four controlled ways:

1. A `Mole` object fixes the geometry, real spherical AO basis, contractions, normalization, AO ordering, shell metadata, and AO-to-anchor map. The occupancy-independent one-electron reference does not run RKS.
2. Libcint supplies independent analytic zero-field overlap, kinetic, local nuclear-attraction, position, and canonical-momentum integrals. These are both oracles and the zero-field analytic part of corrected finite-field contractions.
3. PySCF atom-centred grids and AO evaluators supply values and first derivatives in deterministic blocks. Aion applies Wilson triangle phases, anchored magnetic vectors, electric line integrals, and source variations before contracting the grid blocks.
4. Exact finite-field results use “analytic zero-field plus real-space finite-field correction”. This prevents the complete matrix from inheriting the absolute zero-field quadrature error while retaining an independent raw-grid route for identity checks.

The same blocked contractions run with NumPy or CuPy arrays. Backend residency is explicit; physical GPU tests verify that the CuPy path does not silently fall back to host arrays. GPU evidence applies to the qualified matrix/propagation paths, not to an unimplemented nonlinear TDDFT closure.

5.2 Reusable code map

Table 2: Primary reusable implementation files.

File	Responsibility
<code>src/aion/electronic_structure/magnetic_matrices.py</code>	Exact static uniform-magnetic overlap, kinetic sectors, nuclear attraction, mechanical matrices, analytic first derivatives, direct-gauge oracle, and approximation hierarchy.
<code>src/aion/electronic_structure/time_connection.py</code>	Maxwell-consistent source samples; exact factorized and direct temporal connection; independently differentiated metric rate; electric, magnetic, and weak vector-potential source directions.
<code>src/aion/formulations/exact_one_electron.py</code>	EX/P0/E1/gB1/B1/C1 EOM triples; source directions; pure-gauge response; site charge, oriented-link current, continuity, electric power, and weak current pairings.
<code>src/aion/formulations/action.py</code>	Restricted one-electron action value and fixed-history, history, and full directional derivatives, with resolved metric, connection, and mechanical contractions.
<code>src/aion/propagation/tensorial.py</code>	Mixed/contravariant density conversion, complete generator, fourth-order Gauss–Magnus/[2/2]-Padé link, similarity transport, and geometric diagnostics.
<code>src/aion/electromagnetism/test_variations.py</code>	Smooth localized Gaussian vector-potential variations and their straight-line integrals for weak continuum-current pairings.
<code>src/aion/io/transaction.py</code>	Transactional artifact writing used by authenticated campaign runners.

The runner, analyzer, verifier, fixture, and test inventories are listed in Appendices E and F. Those scripts are campaign infrastructure; the physics and algorithms above live in reusable `src/aion` modules.

6 Evidence architecture and numerical error policy

Each raw execution records the repository branch/commit/dirty state, dependency versions, molecule and basis, AO metadata, grid policy and fingerprint, electromagnetic representative and origin, model level, precision/backend, tolerances, semantic array hashes, and raw residuals. Typical immutable members are `execution_plan.json`, `manifest.json`, `result.json`, `arrays.npz`, and `completed.json`. Derived analyses contain their own authenticated summary, CSV tables, figures, and source hashes.

Numerical conclusions retain three separate scales:

- the algebraic/roundoff residual;
- the real-space quadrature or basis refinement change; and
- the physical model-minus-reference difference.

A physical difference is not credited unless it is separated from the appropriate numerical scale. Field “thresholds” are sampled brackets: the last tested point below a declared relative error and the first tested point above it. No interpolation or universal cutoff is implied.

7 Static qualification: G0–G4

7.1 G0–G1: zero-field reference and quadrature floors

The initial H–H/STO-3G checkpoint reconstructed overlap and weak-form kinetic matrices on unpruned PySCF grids. At grid level 4 (52,080 points), relative Frobenius residuals against libcint were 1.5103×10^{-11} for overlap and 2.8208×10^{-10} for kinetic energy. G1 extended the test to O–H, nuclear attraction, total mechanical energy, canonical momentum, levels 0–5, and NWChem pruning.

The accepted default is unpruned level 4; unpruned level 5 is the refinement; NWChem-pruned level 4 is an alternative-pruning check. The largest measured level-5 working floor was the O–H canonical-momentum residual, 2.21×10^{-8} , still roughly 450 times below the declared 10^{-5} grid target. Momentum-sign discrimination exceeded 9.4×10^7 for O–H and 5.18×10^9 for H–H at level 4.

7.2 G2: exact static Wilson evaluator

For H–H/STO-3G at $|B| = 0.08$ a.u., the maximum EX-direct/EX-link residual was 5.70×10^{-16} . Exact Gram matrices remained positive, with minimum eigenvalue 0.3406818. Symmetric/Landau representatives and two gauge origins gave generalized-spectrum shifts no larger than 2.78×10^{-16} Ha. The O–H rigid-rotation fixture, including oxygen *p*-orbital mixing, gave a maximum generalized-spectrum change of 2.42×10^{-11} Ha, well below the declared propagated tolerance.

7.3 G3–G4: pair scan, first-order hierarchy, and channel attribution

The accepted pair campaign contains 4 systems, 3 bases, 3 bond scales, 4 directions, both field signs where required, and field magnitudes from 10^{-7} to 1 a.u. The exact field data then serve as the parent for the analytic first magnetic derivatives and finite-field remainder fits.

Figure 1 is the compact global view. At relative error 10^{-4} , P0 is conservatively confirmed only through 10^{-4} a.u. for all four matrix families. gB1 extends overlap to 3.162×10^{-3} a.u. but leaves the other families at 10^{-4} a.u. B1/C1 extend overlap, kinetic, and mechanical to 3.162×10^{-3} a.u. and nuclear attraction to 10^{-2} a.u. At the 10^{-2} error level the corresponding conservative brackets are 10^{-2} for P0; 3.162×10^{-2} for gB1 overlap; and 3.162×10^{-2} for B1/C1 overlap, kinetic, and mechanical, with nuclear attraction confirmed through 0.1 a.u.

The finite-field fits in Figure 2 show the intended hierarchy: P0 errors are first order where the first internal amplitude is missing; gB1 is second order only for overlap; B1 and C1 are second order for all static magnetic matrix families. C1 equals B1 here because the static electric sector is absent.

The resolved mechanical audit finds 1,502 blocks that require the anchored- vector B1 contribution: 372 onsite and 1,130 offsite. Across angular blocks, the minimum, median, and maximum B1 error-reduction factors are 11.65, 183.10, and 868.36. The independent uniform-electric E1 coefficient agrees with its exact parent to 1.36×10^{-14} ; the largest level-5 position-quadrature relative residual is 7.36×10^{-7} .

8 Multicentre loop flux and spectra: G5

Two three-proton geometries, two basis sets, and sub-, near-, and above-threshold positive-metric fields test the feature absent from a single pair: a nonzero centre-loop holonomy. The exact gauge-spectrum residual is at most 5.80×10^{-15} Ha and the loop identity residual is at most 8.67×10^{-16} . No metric regularization is used.

At the above-threshold cc-pVDZ point, the distorted geometry gives maximum generalized-eigenvalue shifts of 14.203 Ha (P0), 1.0509 Ha (gB1), and 0.04216 Ha (B1/C1). The associated largest subspace angles are approximately $\pi/2$, $\pi/2$, and 0.02590 rad. The equilateral values are

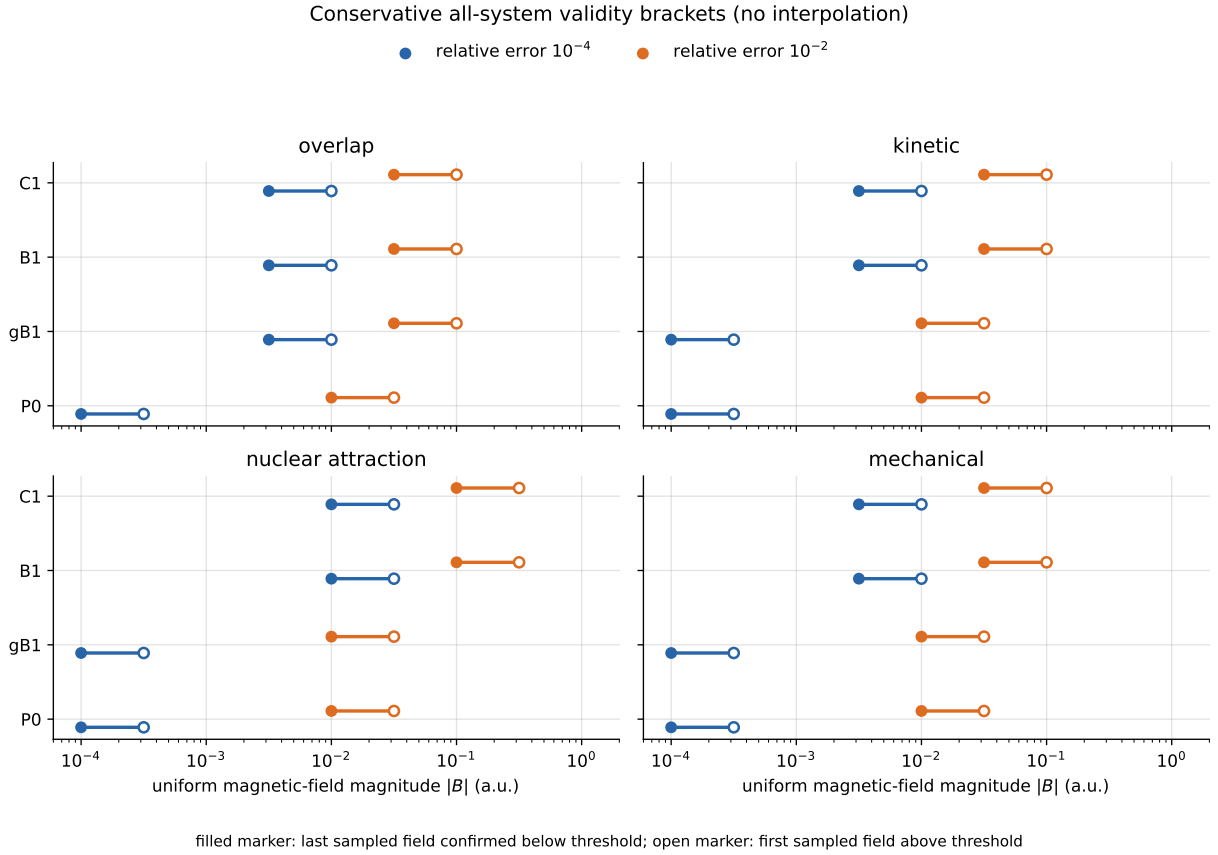


Figure 1: Conservative all-system scan brackets from the authenticated WP8 synthesis. A filled point is the last sampled field below the indicated relative-error threshold; an open point is the first sampled field above it. The segment is a bracket, not an interpolated transition.

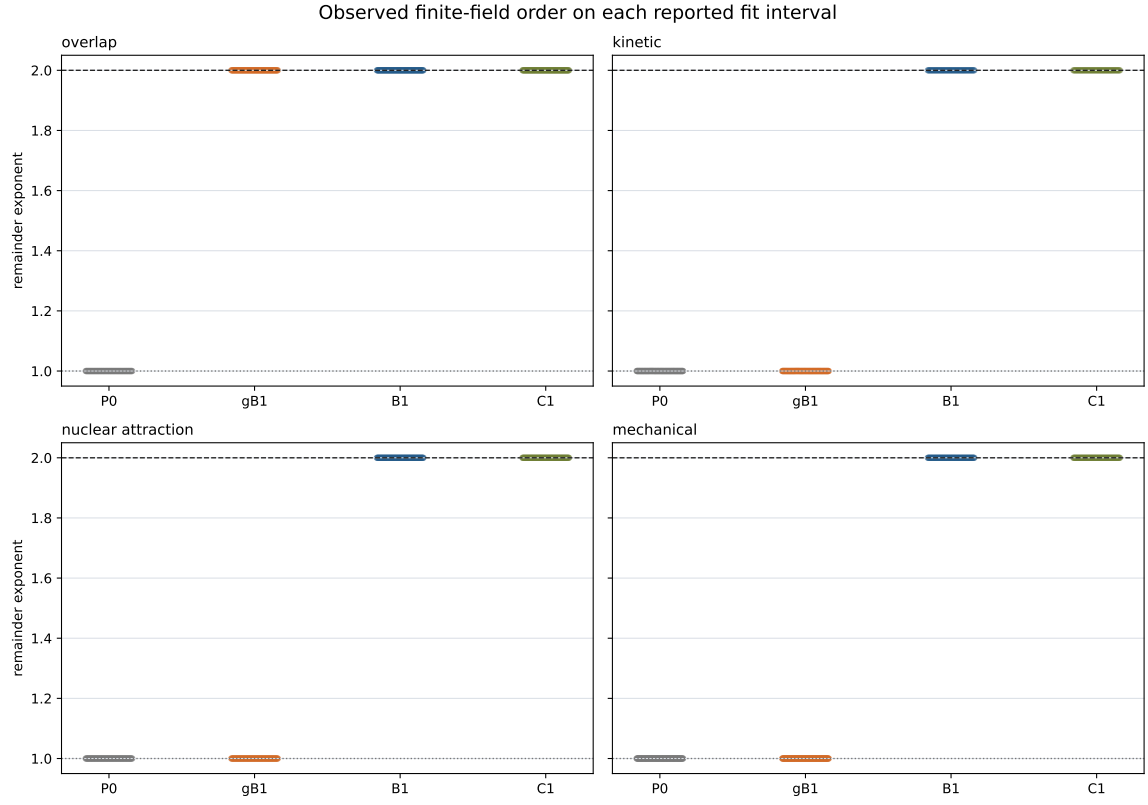


Figure 2: Observed finite-field remainder orders in the WP4 pair campaign. The analytic first-order models remove the corresponding linear term, leaving a quadratic remainder in their declared sectors.

18.260 Ha, 1.6768 Ha, and 0.03184 Ha, with B1/C1 subspace angle 0.00656 rad. Figure 3 shows the complete selected-field trend.

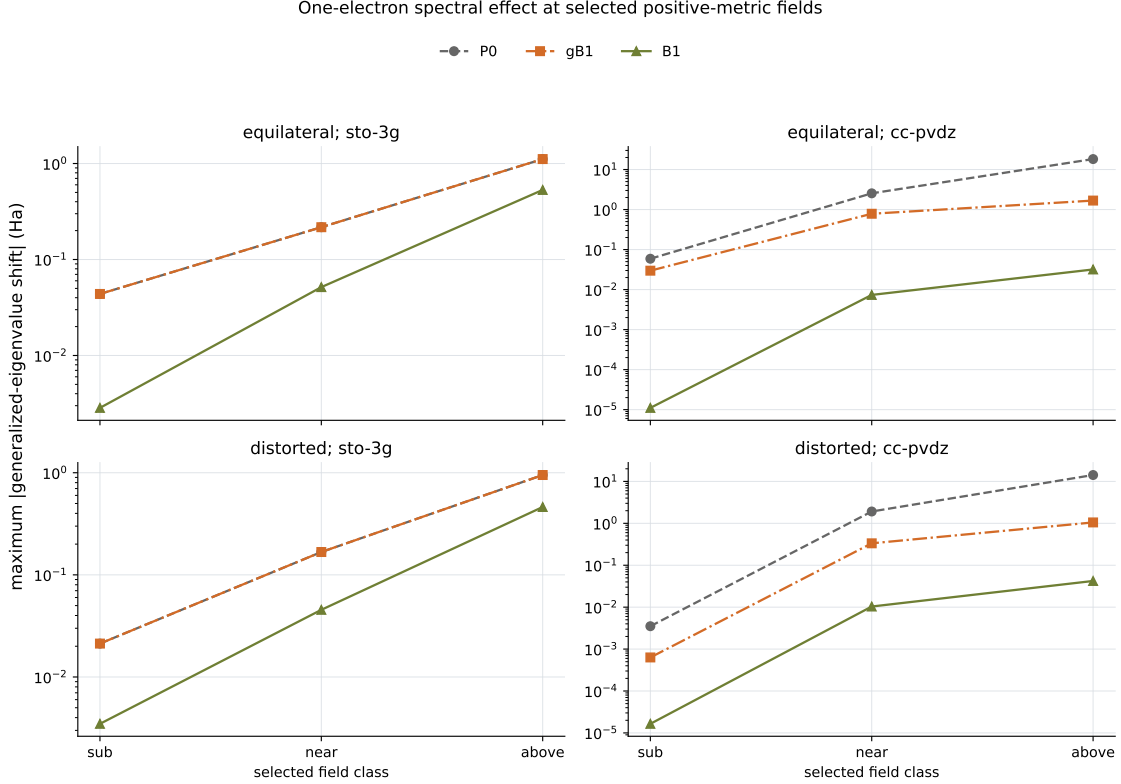


Figure 3: Maximum generalized-eigenvalue shifts relative to EX in the multicentre campaign. The cc-pVDZ panels resolve the strong loop-flux failure of P0/gB1 and the improvement from the full B1 mechanical response.

9 Time-dependent source and connection

For a uniform time-dependent magnetic field, the symmetric representative is

$$\mathbf{A}_{\text{sym}}(\mathbf{r}, t) = \frac{1}{2}\mathbf{B}(t) \times (\mathbf{r} - \mathbf{o}), \quad \Phi_{\text{sym}}(\mathbf{r}, t) = -\mathbf{E}_o(t) \cdot (\mathbf{r} - \mathbf{o}). \quad (16)$$

It produces

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_o(t) + \frac{1}{2}(\mathbf{r} - \mathbf{o}) \times \dot{\mathbf{B}}(t). \quad (17)$$

The affine induction term is compulsory when $\dot{\mathbf{B}} \neq 0$. Landau gauge adds the corresponding $-\partial_t \lambda$ to the scalar potential, so both representatives describe the same $\mathbf{E}$ and $\mathbf{B}$.

The exact endpoint-factorized connection is evaluated independently of a direct-grid differentiation of the ket Wilson factor. The metric rate is assembled by differentiating both the endpoint link and triangle factor, then checked against both $\omega_t + \omega_t^\dagger$ and a central finite difference of independently evaluated static metrics. The connection is therefore known analytically at every requested source time; no auxiliary ODE for S or ω_t is integrated.

10 Time integration: exploration and accepted algorithm

10.1 Why the original checkpoint failed

The original WP6 checkpoint evaluated one complete EOM triple at each midpoint, advanced coefficients with a diagonal [2/2] Padé step, and applied a right-Cholesky endpoint correction.

Its 8/16/32 supplement showed second-order trajectory convergence (minimum measured order 1.9855), good norm and metric-compatibility residuals, and a positive metric. Nevertheless, the maximum 16-to-32 change in a model-minus-exact density difference was 3.338 times the final difference, above the predeclared 0.25 limit.

This was not evidence that S became singular: the minimum metric eigenvalue was 6.54×10^{-3} . It showed that the observable used to compare models was still dominated by propagation error. The result is preserved as failed scientific evidence rather than hidden or reclassified.

10.2 Mixed-index geometry

Define the natural mixed density

$$D^\mu{}_\nu = P^{\mu\lambda} S_{\lambda\nu} = PS. \quad (18)$$

With

$$G = S^{-1} \left(-\omega_t - \frac{i}{\hbar} K \right), \quad (19)$$

metric compatibility converts the contravariant-density equation into

$$\boxed{\dot{D} = [G, D]}. \quad (20)$$

If U solves $\dot{U} = GU$, then

$$D(t_1) = U(t_1, t_0) D(t_0) U(t_1, t_0)^{-1}. \quad (21)$$

The similarity update preserves trace, eigenvalues, and $D^2 = D$ algebraically. Physical Hermiticity is $D^\dagger S = SD$. The contravariant density is recovered by solving $P = DS^{-1}$, without forming an explicit inverse.

This is the geometric simplification: the time-dependent metric has not vanished, but its connection is contained in the complete generator G , and the mixed-index object evolves by a standard similarity commutator.

10.3 Fourth-order Gauss–Magnus and diagonal Padé link

For a step h , evaluate the complete generator at

$$t_\pm = t_n + \left(\frac{1}{2} \pm \frac{\sqrt{3}}{6} \right) h. \quad (22)$$

The fourth-order Magnus exponent is

$$\Omega_4 = \frac{h}{2} (G_- + G_+) - \frac{\sqrt{3}h^2}{12} [G_-, G_+]. \quad (23)$$

The transport link is the diagonal Padé approximation

$$U_n = R_{22}(\Omega_4) = \left(I - \frac{\Omega_4}{2} + \frac{\Omega_4^2}{12} \right)^{-1} \left(I + \frac{\Omega_4}{2} + \frac{\Omega_4^2}{12} \right). \quad (24)$$

Its truncation begins at fifth order and therefore does not lower the formal fourth order of the Magnus step. The accepted update is $D_{n+1} = U_n D_n U_n^{-1}$.

There is no orthogonalization, endpoint Cholesky factor, or metric correction. Instead Aion records the raw cross-metric defect

$$U_n^\dagger S(t_{n+1}) U_n - S(t_n), \quad (25)$$

as well as trace, idempotency, mixed metric-Hermiticity, recovered- P Hermiticity, generator norms, and generator commutators. The current API is named **Experimental** because it consumes a prequalified linear history; the self-consistent nonlinear Gauss-node solve is not yet implemented.

10.4 Accepted convergence and stability

The targeted 32/64/128 continuation of the hardest selected fixture gives global orders 3.978 (EX), 3.707 (P0), 3.716 (E1), 3.986 (gB1), 3.990 (B1), and 3.978 (C1). P0 and E1 require finer steps because their approximate generators are larger and more strongly noncommuting; the eventual fourth-order regime demonstrates a pre-asymptotic timestep effect, not a failure of the mixed-index equation.

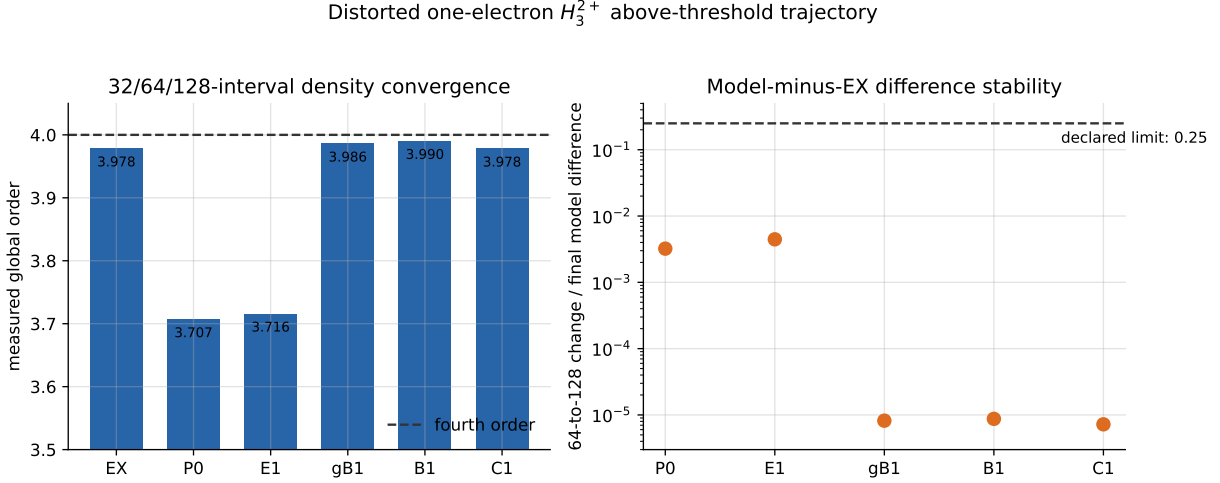


Figure 4: Final hard-case convergence evidence. The left panel isolates the 32/64/128 measured order for each model. The right panel compares the 64-to-128 change in each model-minus-EX density difference with the declared 0.25 stability limit.

Across the accepted continuation, trace drift is below 6.7×10^{-16} , the idempotency residual below 5.5×10^{-15} , and the minimum metric eigenvalue is 6.536×10^{-3} .

11 Dynamic model comparison: G6

The distorted above-threshold pulse has duration 4 a.u. and peak centre-loop flux 0.05 a.u. It simultaneously activates the electric and magnetic finite-spread sectors. The field returns to the declared endpoint, so the reported endpoint energy is evaluated with the common field-free mechanical operator.

At 128 intervals ($\Delta t = 0.03125$ a.u.), EX absorbs 2.805397×10^{-5} Ha. The approximate values are 1.781478×10^{-4} (P0), 1.885249×10^{-4} (E1), 4.921313×10^{-4} (gB1), 1.706561×10^{-4} (B1), and 2.783231×10^{-5} Ha (C1). The exact excitation probability is 2.199458×10^{-5} and C1 gives 2.189968×10^{-5} .

The endpoint metric Hilbert–Schmidt density distances from EX are 0.01428 (P0), 0.01025 (E1), 0.02391 (gB1), 0.01457 (B1), and 8.8254×10^{-4} (C1). Figure 5 shows that the observables have stabilized over 16–128 intervals; Figure 6 places the final physical separation beside the 64-to-128 difference uncertainty.

The physical conclusion is not that B1 “fails” generally. It is that a magnetic-only first-order model cannot reproduce a trajectory that also contains the internal electric E1 response. Conversely, E1 has the electric sector but omits the magnetic metric and mechanical sectors. C1 contains both at the same action level.

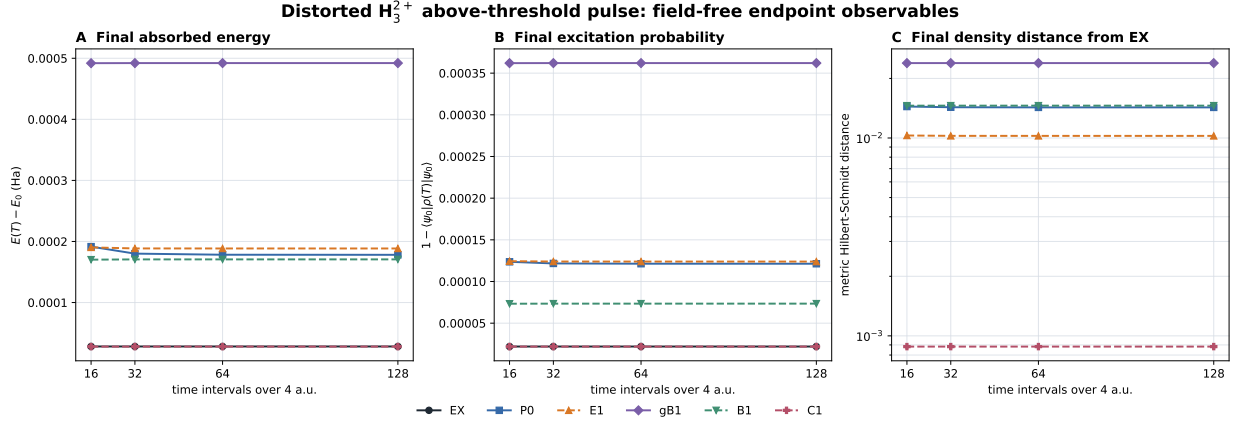


Figure 5: Field-free endpoint observables for the distorted one-electron H_3^{2+} above-threshold pulse. C1 is the only tested first-order model that remains close to EX in all three panels.

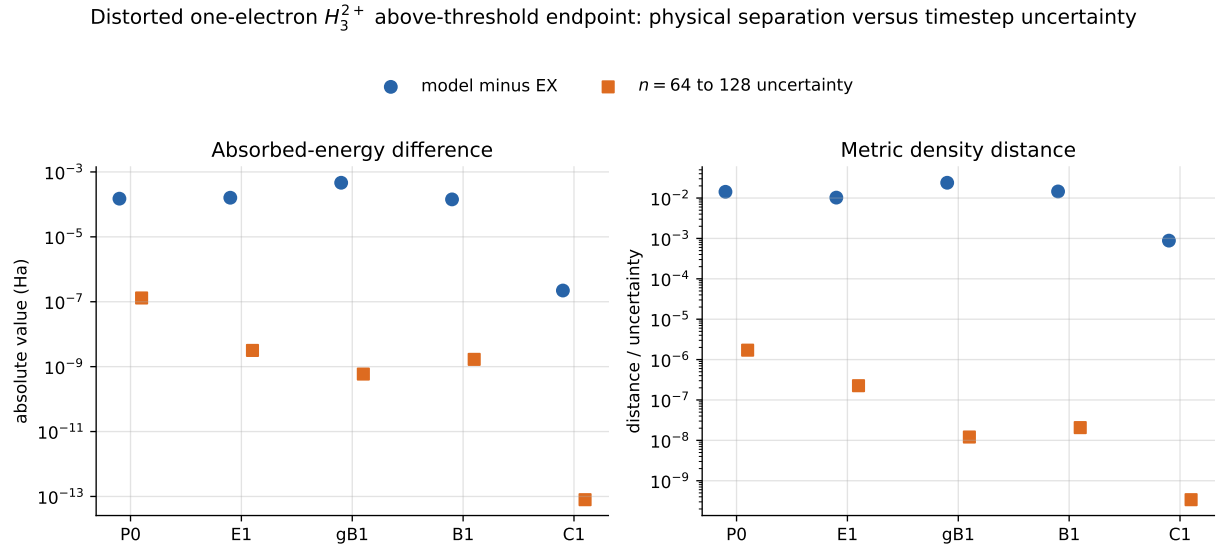


Figure 6: Absolute model-minus-EX endpoint differences compared directly with their 64-to-128 timestep difference uncertainty. The many-decade separation for C1 establishes a resolved truncation error rather than a time-integration artifact.

12 Action observables, continuity, and power: G7

12.1 Variational source directions

For fixed coefficient history, a real source parameter λ with matrix directions $(S_\lambda, K_\lambda, \omega_\lambda)$ gives the implemented instantaneous action derivative

$$\left. \frac{\partial L}{\partial \lambda} \right|_{C, \dot{C}} = \Re \left\{ \frac{i\hbar}{2} \text{Tr}[(R - R^\dagger)S_\lambda + P(\omega_\lambda - \omega_\lambda^\dagger)] - \text{Tr}(PK_\lambda) \right\}, \quad (26)$$

where $R = \dot{C}fC^\dagger = GP$ on shell. Aion separately exposes the metric kinematic, connection kinematic, and mechanical contributions. Currents are therefore derivatives of the same restricted action used for propagation, not ambient momentum expectations borrowed from another model.

The implemented directions include independent anchor scalar potentials, oriented endpoint links, complete instantaneous uniform $\mathbf{B}$, separated endpoint/internal magnetic parts, pure-gauge source/history pairs, uniform electric components, and localized Gaussian vector-potential variations. Site charges and oriented pair currents obey the declared incidence convention $\dot{Q} + \mathcal{I}I = 0$.

The projected ambient mechanical current remains a useful diagnostic, but in a finite source-dependent subspace it need not equal the variational source current. Their difference measures finite-subspace and source-response effects; it is not automatically an implementation error.

12.2 Uniform-electric power

In the zero- B specialization the three electric-field action derivatives define the electronic polarization. Applying the same covectors to $\dot{P}$ defines $\mathbf{J}_{\text{src}}$ and yields

$$\frac{dU}{dt} = \mathbf{E}(t) \cdot \mathbf{J}_{\text{src}}(t). \quad (27)$$

No momentum operator is inserted into this identity. The finest $\Delta t = 0.125$ a.u. trajectory has maximum instantaneous power residual 4.88×10^{-16} , while the integrated work defect is -1.236×10^{-9} Ha. The difference is expected: instantaneous source contraction is algebraic, whereas accumulated work and a centred trajectory derivative carry fourth- and second-order time-discretization errors, respectively.

Measured integrated-work orders are 3.9658 and 3.9914; centred derivative orders are 1.9464 and 1.9931. At the finest electric step, the maximum continuity, Ward, and total-charge residuals are 2.57×10^{-15} , 3.89×10^{-16} , and 1.22×10^{-15} .

12.3 Magnetic identities and GPU parity

For the finest magnetic trajectory, the maximum cross-metric residual is 3.30×10^{-12} ; continuity, Ward, and total-charge residuals are 3.93×10^{-15} , 4.44×10^{-16} , and 6.66×10^{-16} . Figure 9 shows their history.

The physical-GPU n=32 comparison gives maximum CPU/GPU absolute differences of 1.94×10^{-15} (electric density), 3.18×10^{-15} (electric diagnostics), 1.53×10^{-15} (magnetic density), and 4.14×10^{-15} (magnetic diagnostics). The final fast CPU suite before G8 reported 88 passes and 133 deselections; targeted physical-GPU tests were also executed and accepted.

13 G8 synthesis, accepted boundary, and next branch

The WP8 synthesis authenticates the accepted G3–G7 records and produces five compact tables:

- 768 system/basis-specific magnetic validity rows;
- 64 conservative global validity rows;
- 10 mechanical channel-attribution rows;

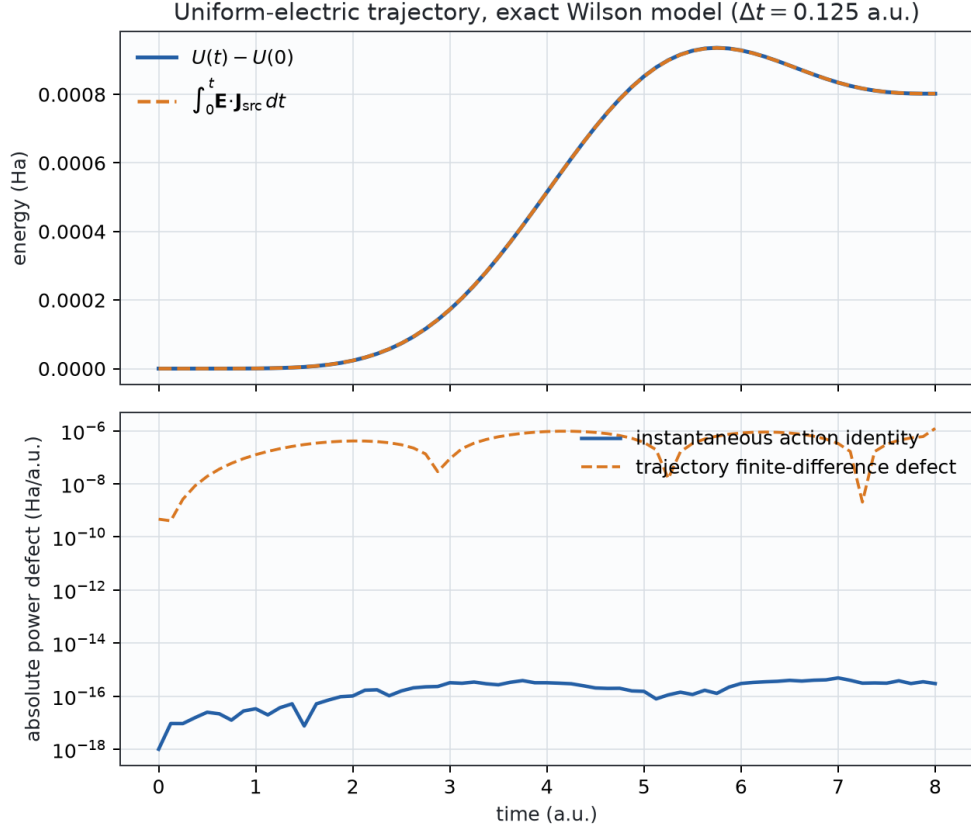


Figure 7: Uniform-electric exact-Wilson trajectory. The upper panel compares mechanical-energy change with integrated source work. The lower panel separates the roundoff-level instantaneous action identity from the expected finite-difference trajectory derivative error.

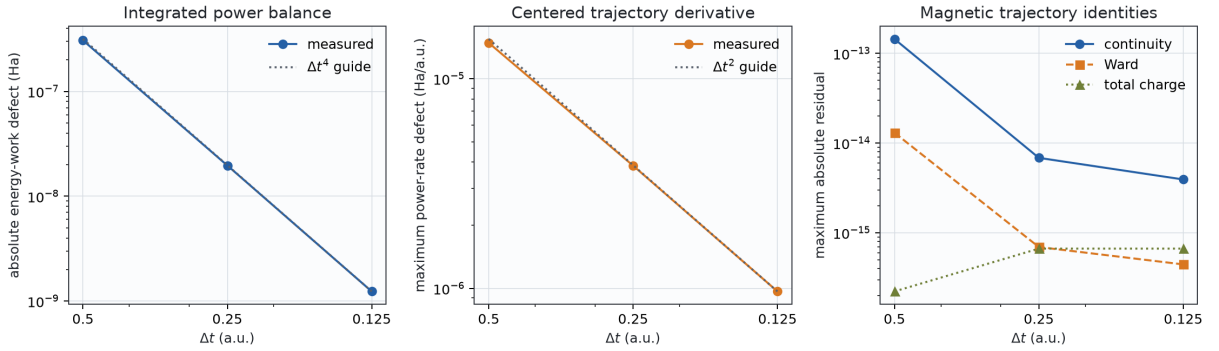


Figure 8: WP7 refinement summary. Integrated power balance converges at fourth order, the centred trajectory derivative at second order, and the continuity, Ward, and charge identities remain near roundoff.

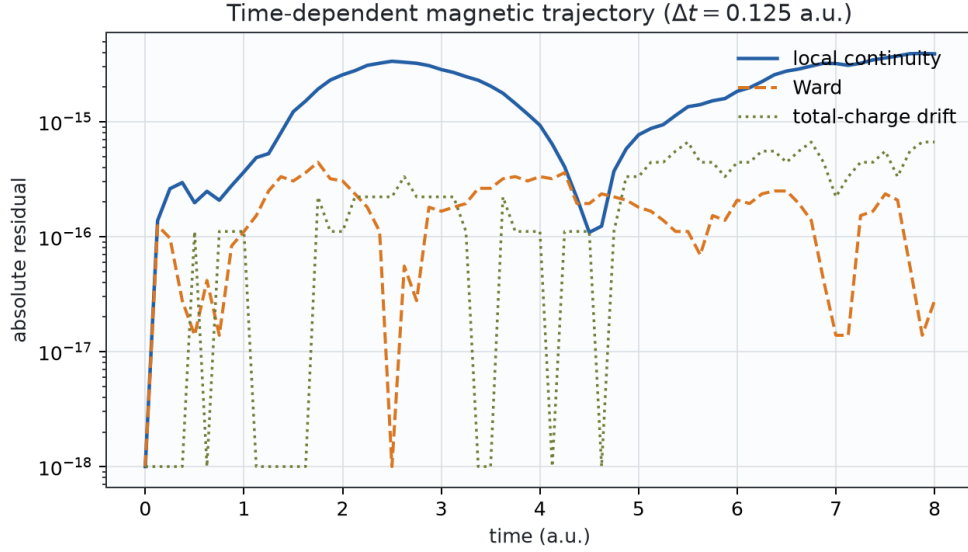


Figure 9: Action-derived local continuity, pure-gauge Ward, and total-charge residuals on the finest time-dependent magnetic trajectory.

- 48 multicentre spectral-comparison rows; and
- 5 dynamic endpoint-difference rows.

G8 was explicitly accepted. The bounded validity statement is:

The fixed-centre linear exact one-electron Wilson dynamics and its P0, E1, gB1, B1, and C1 truncations are qualified only over the explicitly tabulated systems, bases, geometries, fields, source histories, quadrature levels, and timesteps. The reported ranges are scan brackets and every physical comparison retains its quadrature or timestep companion.

The accepted post-G8 choice is *Aion adiabatic pure-LDA/GGA TDDFT closure*. This follows from the measured boundary: source-only exact Wilson geometry, connection-aware linear propagation, action source derivatives, and CPU/GPU parity now have accepted evidence, while self-consistent Hartree/XC response and nonlinear Gauss-node propagation do not. This report does not plan or implement that branch; it closes and indexes the evidence on which the branch choice rests.

A Authoritative documents

The mathematical authority and operational plan are outside the Aion worktree; implementation notes and review decisions are versioned with the code. The following links are intended as the reading order.

1. [Wilson-dressed adiabatic TDDFT reference \(TeX\)](#) — complete fixed-centre action, density, XC/RI-Hartree closure target, source derivatives, and staged realization.
2. [Wilson-dressed adiabatic TDDFT reference \(PDF\)](#).
3. [Exact one-electron numerical qualification plan](#) — gate definitions and original acceptance criteria. Its embedded status table is historical; the review records below are authoritative for final gate state.
4. [Exact one-electron qualification agent brief](#) — execution discipline and evidence-status contract used for the campaign.
5. [Exact Wilson matrix-element qualification note \(TeX\)](#) and [PDF](#).
6. [G0 conventions and schema note](#).
7. [WP1 overlap/kinetic checkpoint](#) and [WP1 zero-field qualification](#).
8. [WP2 static Wilson qualification](#).
9. [WP6 time connection and linear propagation](#).
10. [Mixed-density Gauss-Magnus experiment](#).
11. [WP8 synthesis narrative](#).

Related general Aion documents are:

- [theory and implementation](#);
- [user guide](#);
- [configuration and schema contracts](#); and
- [magnetic benchmark API](#).
- [magnetic-matrix benchmark theory](#);
- [magnetic-matrix benchmark contract](#);
- [magnetic-matrix benchmark implementation plan](#); and
- [Aion refactor implementation plan](#).

B Gate review records

Gate	Decision	Record
G0	accepted	G0 review
G1	accepted	G1 review
G2	accepted	G2 review
G3	accepted	G3 review
G4	accepted	G4 review
G5	accepted	G5 review
G6	accepted	G6 review
G7	accepted	G7 review ; the preserved candidate is pre-decision evidence.
G8	accepted	G8 review ; the preserved candidate contains the pre-decision synthesis and recommendation.

C Campaign directory index

All roots below live under `/home/cgs/00_WORK/Projection_Code/CALCULATIONS/campaigns/exact_one_electron_qualification`. The sizes are approximate and help identify the heavy raw data. The accepted roots are identified by the review records; historical and failed executions remain part of the audit trail.

Directory	Size/files	Purpose and status
20260916T184521Z_3fd2a83b77d1	3.4 MB / 17	Initial G0 checkpoint execution retained as historical duplicate/restart; not the accepted root.
20260916T184810Z_3fd2a83b77d1	3.4 MB / 17	Accepted G0 H–H overlap/kinetic checkpoint; inspect <code>execution_index.json</code> .
wp1_20260916T190311Z_3fd2a83b77d1	24 MB / 74	Initial full WP1 execution retained as historical duplicate/restart.
wp1_20260916T190346Z_3fd2a83b77d1	24 MB / 74	Accepted G1 zero-field campaign; <code>execution_index.json</code> is the entry point.
wp2_20260916T192437Z_3fd2a83b77d1	6.6 MB / 5	Accepted G2 exact static Wilson evaluator/covariance campaign; <code>manifest.json</code> , <code>result.json</code> , <code>arrays.npz</code> .
wp3_pair_20260916T195452Z_3fd2a83b77d1	1.9 MB / 4	Initial WP3 pair-scan start retained for provenance; superseded by the complete scan.
wp3_pair_20260916T195529Z_3fd2a83b77d1	531 MB / 76	Complete reference-distance pair scan and its first analysis; figures/tables are under <code>analysis</code> .
wp3_pair_distance_20260917T002456Z_3fd2a83b77d1	1.1 GB / 123	Accepted G3 distance-complete pair campaign; combined entry point is <code>g3_analysis/report.md</code> .
wp4_first_order_20260918T191425Z_3fd2a83b77d1	35 MB / 167	Accepted G4 analytic first-order hierarchy, channel classification, and E1 audit; use <code>analysis/report.md</code> .
wp5_multicentre_20260918T200808Z_3fd2a83b77d1	11 MB / 34	Accepted G5 loop-flux, metric-domain, generalized-spectrum, and selected-dynamics cases; use <code>analysis/report.md</code> .
wp6_linear_20260918T204830Z_3fd2a83b77d1	328 KB / 5	Smoke run; <code>completed.json</code> says <code>smoke_not_qualification</code> .
wp6_linear_20260918T205152Z_3fd2a83b77d1	356 KB / 5	Second smoke run; <code>smoke_not_qualification</code> .
wp6_linear_qualification_20260918T2055Z_3fd2a83b77d1	2.0 MB / 29	Full initial midpoint/Cholesky qualification. Important source evidence, but its model-difference stability was unresolved.
wp6_h3_timestep_refinement_20260918T212428Z_3fd2a83b77d1	1.8 MB / 22	Authenticated 8/16/32 supplement: second-order convergence, failed 0.25 model-difference stability criterion.
wp6_mixed_magnus_20260918T231057Z_38ce83eb2fe9	16 MB / 16	Replacement mixed-index Gauss–Magnus experiment and derived convergence/stability analysis.
wp6_mixed_magnus_n64_20260919T120325Z_57ef2fb6d40e	3.0 MB / 10	Targeted 64-interval continuation and endpoint observables.
wp6_mixed_magnus_n128_diagnostics_20260919T124656Z_73f6dc1cb760	5.7 MB / 10	Accepted hard-case 128-interval continuation, full diagnostics, convergence, stability, and endpoint observables.
wp7_observables_cpu_20260919T145221Z_7d603105e1db	496 KB / 10	Accepted CPU action-observable histories, refinement table, figures, and summary.

Directory	Size/files	Purpose and status
wp7_observables_gpu_ 20260919T145459Z_7d603105e1db	28 KB / 5	Accepted physical-CuPy parity execution.
wp8_synthesis_20260919T152644Z_ 2fa0214fa659	208 KB / 9	Authenticated G8 synthesis, five compact tables, summary, manifest, and narrative.

D Derived tables and figures

The compact final tables are:

- [magnetic validity by system and basis](#);
- [global magnetic validity](#);
- [channel attribution](#);
- [multicentre spectral differences](#); and
- [dynamic endpoint differences](#).

The most useful gate-specific derived roots are:

- [G3 combined analysis](#);
- [G4 analysis](#);
- [G5 analysis](#);
- [failed midpoint supplement analysis](#);
- [mixed-Magnus analysis](#);
- [128-interval endpoint observables](#); and
- [G7 observable analysis](#).

The report-specific figures in `docs/figures/exact_one_electron_qualification` are regenerated by `tools/build_exact_one_electron_qualification_report_assets.py` from the authenticated WP8 tables and WP6 n=128 result. They do not alter the source evidence.

E Runner, analyzer, and verifier index

The campaign tools are grouped by role rather than copied into the scientific API.

Run • `run_exact_one_electron_wp1_checkpoint.py`

- `run_exact_one_electron_wp1_full.py`
- `run_exact_one_electron_wp2.py`
- `run_exact_one_electron_wp3_pair_scan.py`
- `run_exact_one_electron_wp3_reference_audit.py`
- `run_exact_one_electron_wp4_first_order.py`
- `run_exact_one_electron_wp4_e1_audit.py`
- `run_exact_one_electron_wp5_multicentre.py`
- `run_exact_one_electron_wp6_linear.py`
- `run_exact_one_electron_wp6_h3_timestep_refinement.py`
- `run_exact_one_electron_wp6_mixed_magnus.py`
- `run_exact_one_electron_wp6_mixed_magnus_n64.py`
- `run_exact_one_electron_wp6_mixed_magnus_n128_diagnostics.py`
- `run_exact_one_electron_wp7_observables.py`

Analyze

The correspondingly named `analyze_exact_one_electron_*` tools for G3, G4, G5, the WP6 primary/supplement/mixed/endpoint paths, G7, and G8. `analyze_exact_one_electron_wp8_synthesis.py` is the final cross-gate reducer.

Verify

The correspondingly named `verify_exact_one_electron_*` tools independently recompute

manifest identities, hashes, row counts, and declared residual decisions. The WP8 verifier authenticates all five synthesis tables.

All are under `/home/cgs/00_WORK/Projection_Code/aion-magnetic-matrix-benchmark/tools`.

F Fixtures and automated tests

The immutable fixture directory is `tests/fixtures/exact_one_electron`. It contains H–H and O–H reference fixtures, the G2 oblique regression, the WP3 pair campaign, the WP5 multicentre campaign, the WP6 linear/midpoint/mixed/n64/n128 declarations, and the WP7 observables declaration.

The principal CPU tests are:

- `tests/test_exact_one_electron_wp0_wp1.py`
- `tests/test_exact_one_electron_wp1_full.py`
- `tests/test_exact_one_electron_wp2.py`
- `tests/test_exact_one_electron_wp3_analysis.py`
- `tests/test_exact_one_electron_wp4.py`
- `tests/test_exact_one_electron_wp6.py`
- `tests/test_exact_one_electron_wp7.py`
- `tests/test_tensorial_magnus.py`

Physical-GPU tests are:

- `tests/test_exact_one_electron_wp6_gpu.py`
- `tests/test_exact_one_electron_wp7_gpu.py`
- `tests/test_tensorial_magnus_gpu.py`

G Authentication anchors

The accepted WP8 synthesis root has manifest ID

`f41bbd648891028d9665caf13f5ab1801b46c8d87eb535db3a5471c0f88f41c8`.

Its principal SHA-256 values are:

Artifact	SHA-256
<code>completed.json</code>	<code>da2de4de08498cbfe69ce12448f8dc28a5069bd2866d2dc6cc5b8630445bc72d</code>
<code>manifest.json</code>	<code>929ba2f569b824494f3a43c3fb1773d286f2ae818f1611b0e04fe20980420f81</code>
<code>summary.json</code>	<code>4c5cb2146cf63390424ec1c8aa28c1d39d5827f6d146169a9db80b646b9f662c</code>
<code>synthesis.md</code>	<code>e7f31b688e4e75f6e787828ecd073345fe6eafd98bca101129d52688d1ce8bc2</code>

Earlier per-gate manifest IDs and SHA-256 anchors are retained in the G0–G7 review JSON records rather than duplicated here. This ensures that future edits to the narrative do not silently redefine the accepted raw evidence.